

HARI VIGNESH RAO 23BD1A052Q

WEEK-10--Elasticbeanstalk

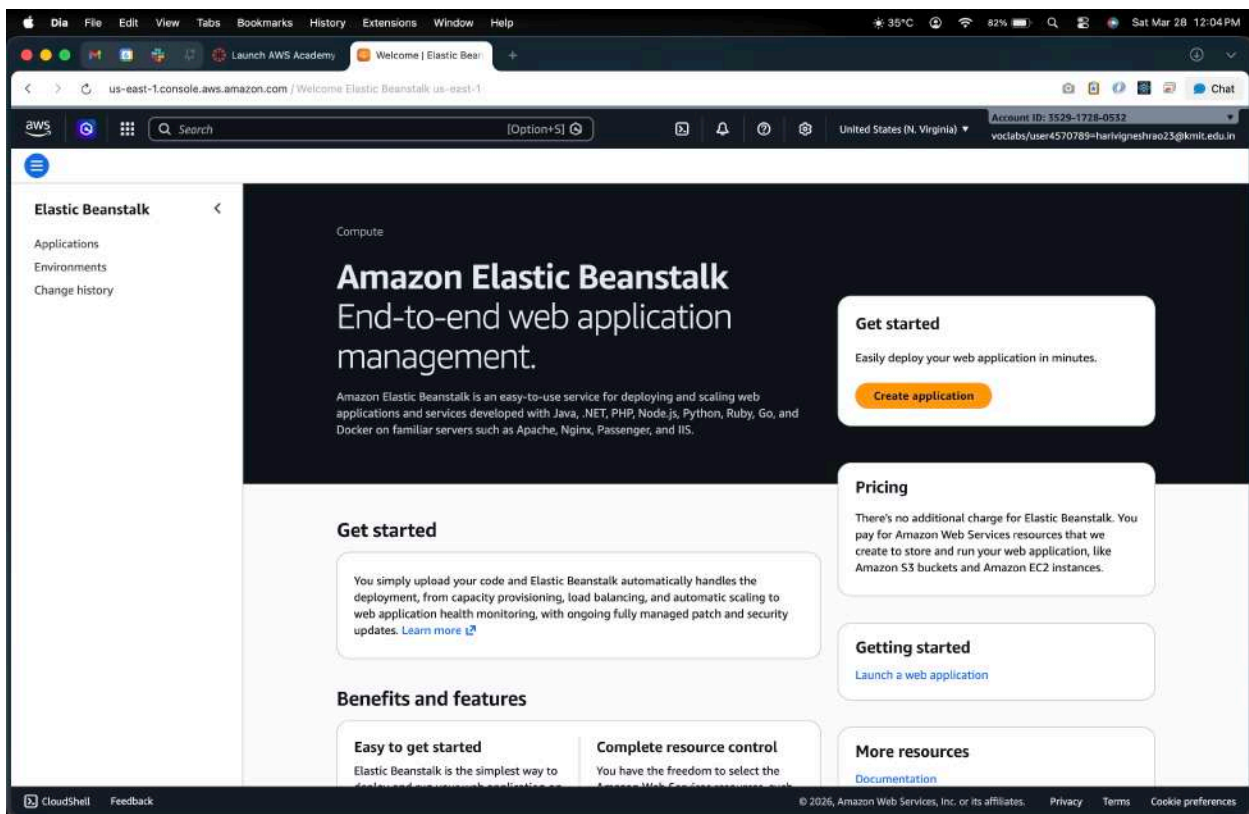
Prerequisites

- A web application ready (e.g., .war, .zip, or code)

Steps to Create Elastic Beanstalk Application

Step 1: Open Elastic Beanstalk

1. Login to AWS Console
2. Under Compute section ->Search for **Elastic Beanstalk**
3. Click **Create Application**



Step 2: Create Application

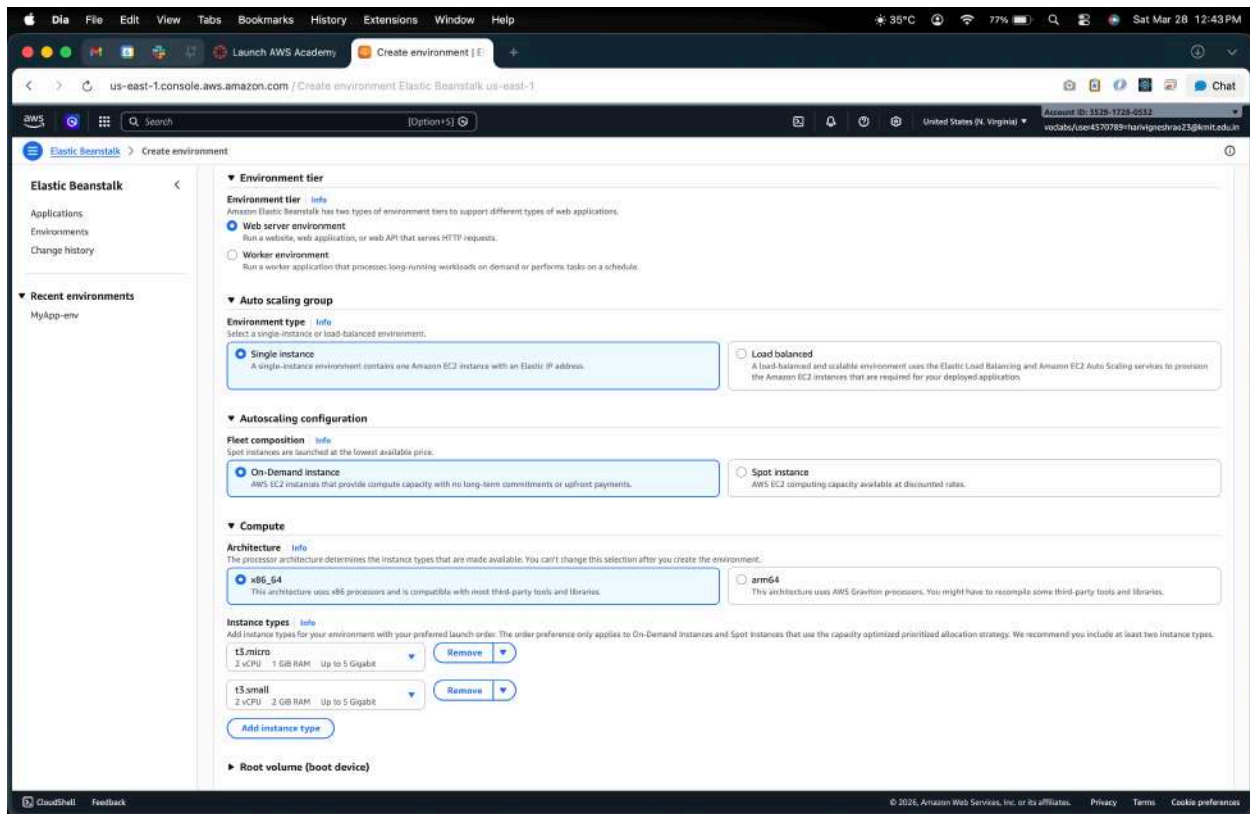
1. Enter **Application Name** (e.g., MyApp)

2. Add description (optional)
3. Click **Create**

The screenshot shows the AWS Elastic Beanstalk 'Create environment' console page. The page is titled 'Create environment' and shows various configuration options for a new environment. The 'Basic details' section includes fields for 'Application name' (MyApp), 'Domain name prefix' (default), 'Environment name' (MyApp-env-1), and 'Environment description'. The 'Platform details' section shows 'Platform' as Tomcat, 'Platform branch' as Tomcat 11 with Corretto 25 running on 64bit Amazon Linux 2023, and 'Platform version' as 5.12.0 (Recommended). The 'Application code' section has 'Local file' selected, with a file named 'HospitalMgmtSystem.war' chosen. The 'Version label' section has a label field. The bottom of the page shows 'Additional environment details - optional'.

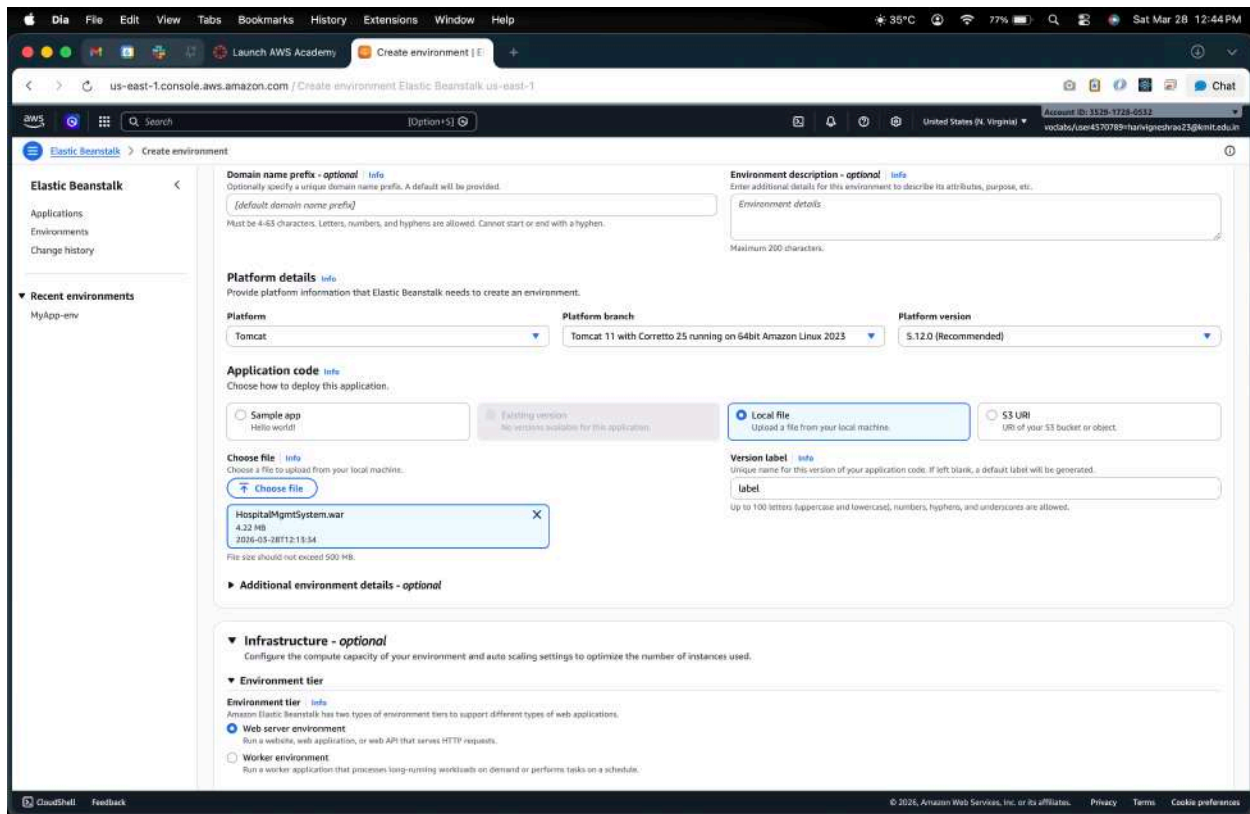
Step 3: Create Environment

1. Click **Create Environment**
2. Choose:
 - **Web Server Environment** (for web apps)
3. Click **Select**



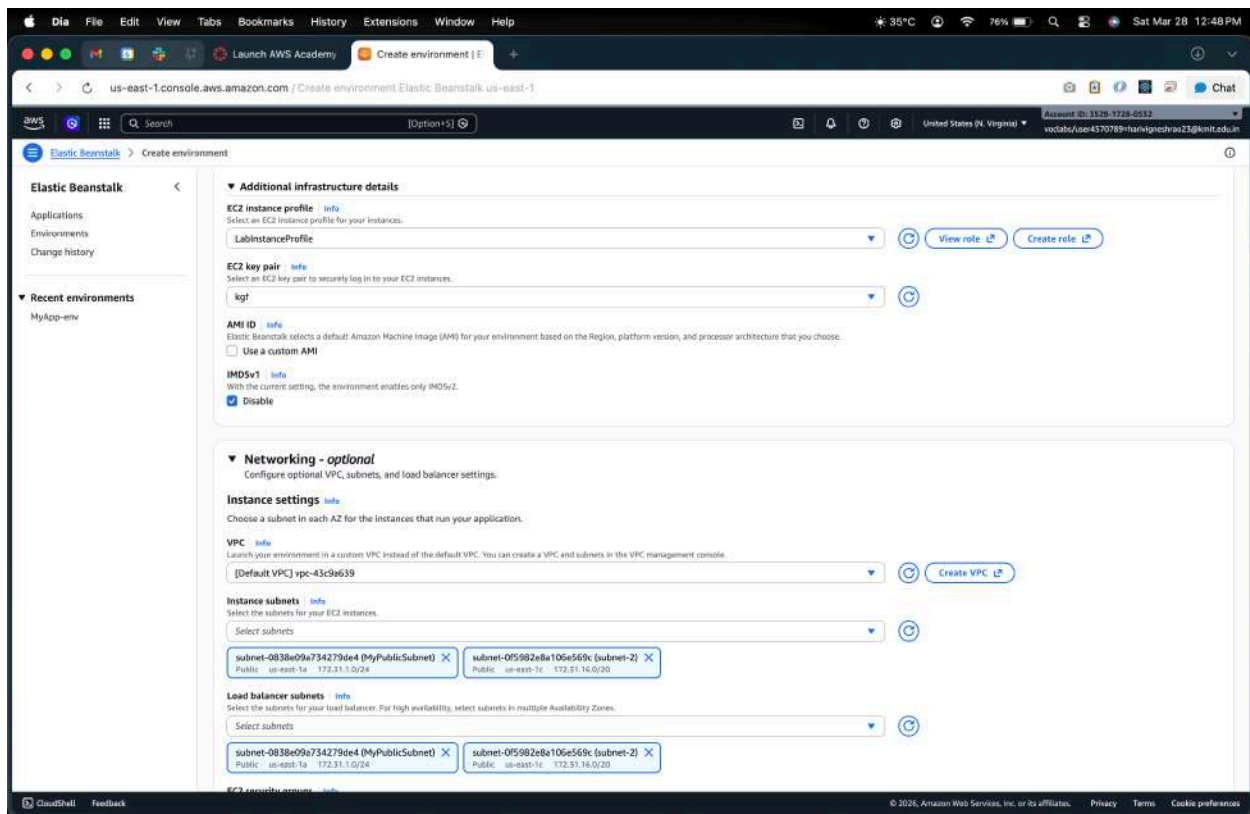
Step 4: Configure Environment

1. Enter **Environment Name**
2. Choose **Platform**:
 - tomcat (for .war)
 - tomcat /Node.js / Python / PHP / etc.
3. Under **Application Code**:
 - Select **Upload your code**
 - Upload your .war or .zip file



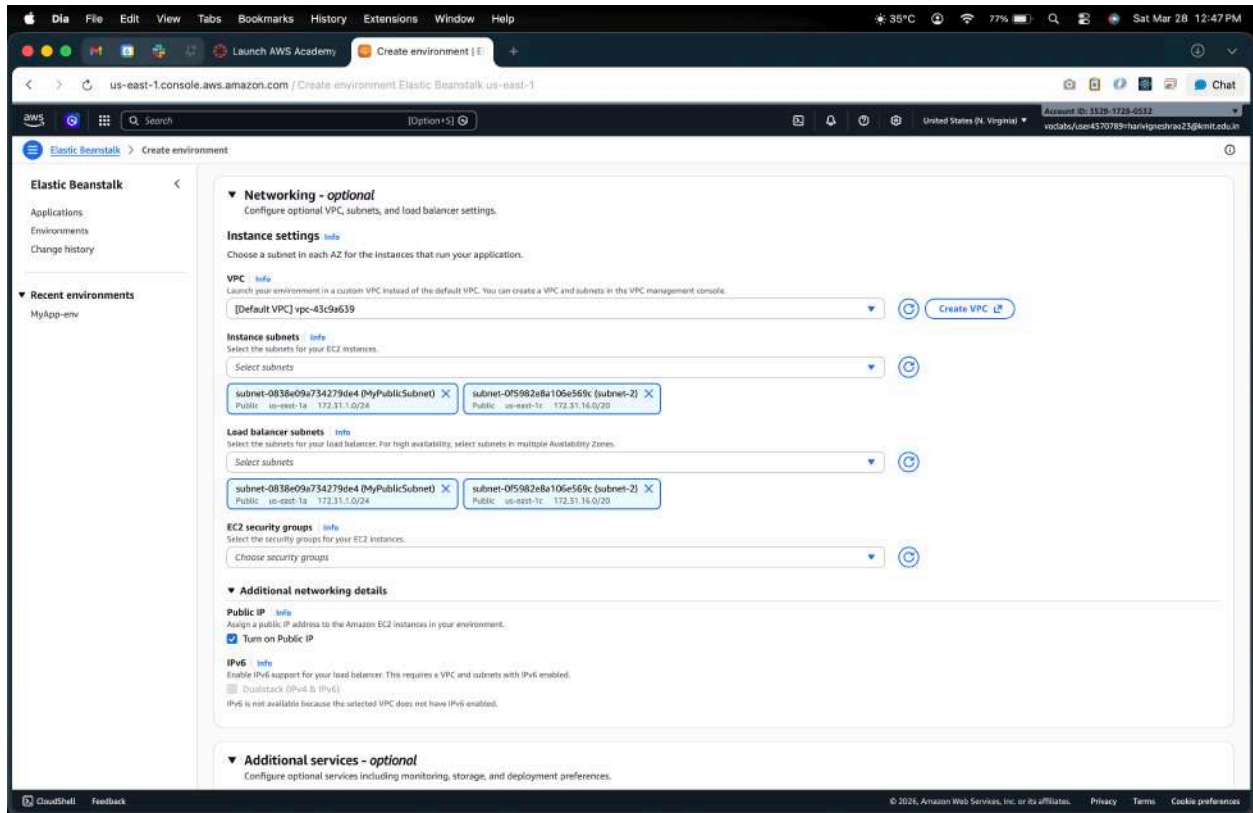
Step 5: Configure Service Access

1. Service Role → Select existing → **LabRole**
2. EC2 Key Pair → Optional (for SSH)
3. EC2 Instance Profile → select (e.g., LabInstanceProfile)



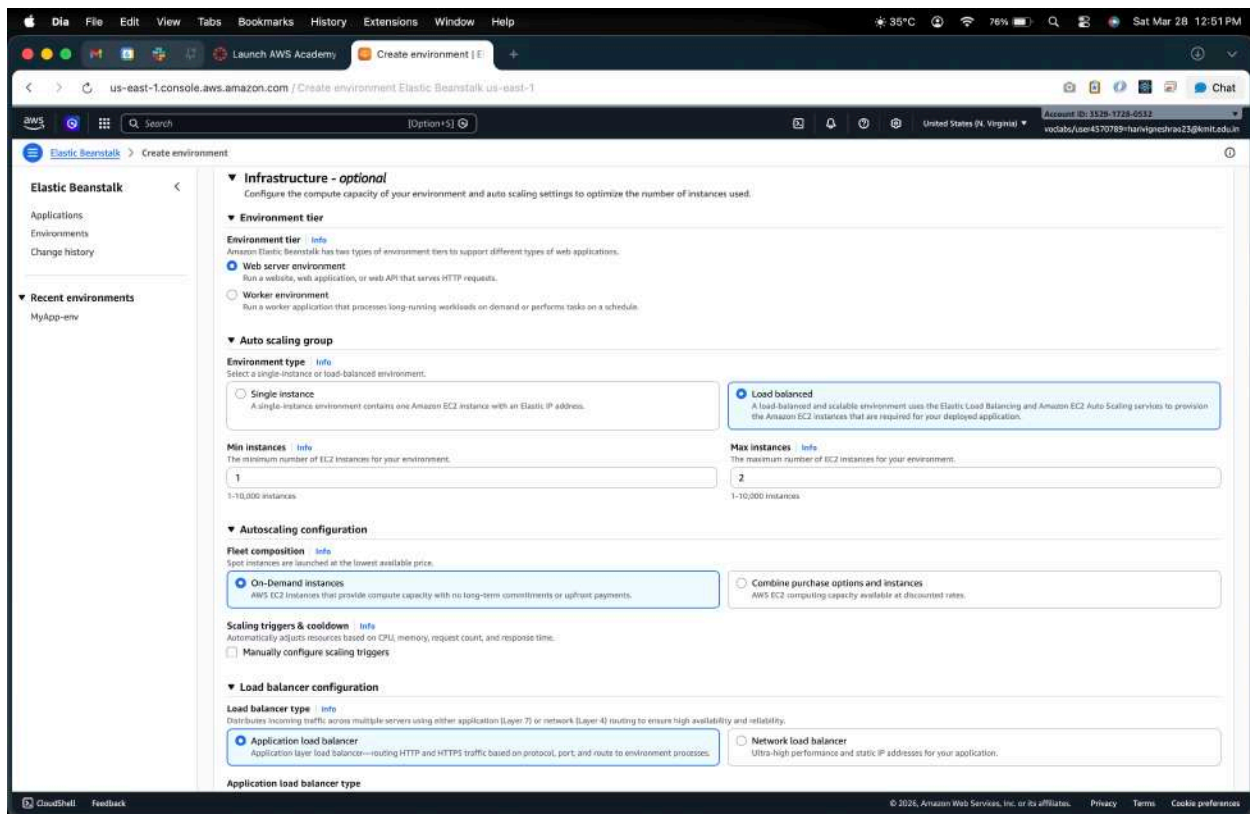
Step 6: Configure Network

1. Select **VPC** (default is fine)
2. Select **Subnets**
3. Enable **Public IP**



Step 7: Instance & Scaling Setup

1. Choose **Instance type** (e.g., t2.micro – free tier)
2. Set:
 - Min instances: 1
 - Max instances: 2 (or more)
3. Load Balancer → Enabled (optional but recommended)

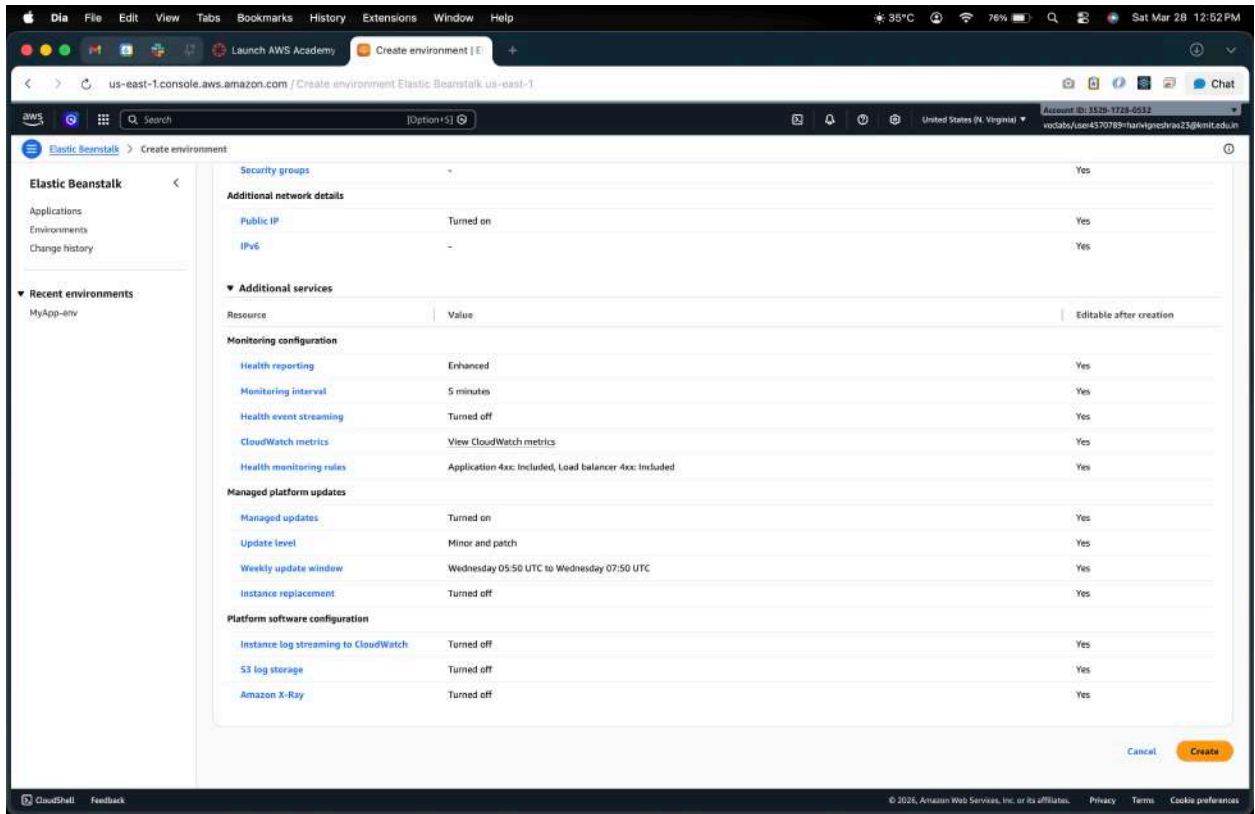


Step 8: Review & Create

1. Click **Review**
2. Click **Create**

The screenshot shows the AWS Elastic Beanstalk console in the 'Create environment' step. The 'Review' section is active, displaying a summary of the configuration details for the application 'MyApp'. The details are organized into several sections: Environment details, Basic details, Platform details, Application code, Additional environment details, and Infrastructure. Each section contains a table of resources and their values, with a column indicating if the value is editable after creation.

Resource	Value	Editable after creation
Basic details		
Application name	MyApp	No
Environment name	MyApp-env-1	No
Domain name	[default domain name]	Yes
Environment description	-	Yes
Platform details		
Platform	Tomcat	Yes
Platform branch	Tomcat 11 with Corretto 25 running on 64bit Amazon Linux 2023	Yes
Platform version	5.12.0	Yes
Application code		
Deployment method	Local file	Yes
Version label	[auto-generated]	Yes
Additional environment details		
Service role	arn:aws:iam::352917280532:role/LabRole	Yes
Environment properties	View properties	Yes
Tags	View tags	Yes
Infrastructure		
Environment tier		
Tier	Web server environment	No



Deployment Process

- AWS automatically creates:
 - EC2 instances
 - Load Balancer
 - Auto Scaling Group
 - Security Groups

us-east-1.console.aws.amazon.com / Environment overview - events Elastic Beanstalk us-east-1

Elastic Beanstalk > Environments > MyApp-env-1

Elastic Beanstalk is launching your environment. This will take a few minutes.

Events (14)

Time	Type	Details
March 28, 2026 12:55:45 (UTC+5:30)	INFO	Created Load Balancer listener named: arn:aws:elasticloadbalancing:us-east-1:352917280532:listener/app/awseb-NvpYdZ0rZ4hw/b6f91e613a7de9f1/bb0ef78b1bb694ba
March 28, 2026 12:55:42 (UTC+5:30)	INFO	Created load balancer named: arn:aws:elasticloadbalancing:us-east-1:352917280532:loadbalancer/app/awseb-NvpYdZ0rZ4hw/b6f91e613a7de9f1
March 28, 2026 12:54:41 (UTC+5:30)	INFO	Created CloudWatch alarm named: awseb-e-g39cncr8-stack-AWSEBCloudwatchAlarmHigh-31y2I7HnE9
March 28, 2026 12:54:41 (UTC+5:30)	INFO	Created CloudWatch alarm named: awseb-e-g39cncr8-stack-AWSEBCloudwatchAlarmLow-8pC3oUnuVnEA
March 28, 2026 12:54:26 (UTC+5:30)	INFO	Created Auto Scaling group policy named: arn:aws:autoscaling:us-east-1:352917280532:scalingPolicy:709a26b7-4df6-4964-a4e6-93c74e19f3c:autoScalingGroupName/awseb-e-g39cncr8-stack-AWSEBAutoScalingGroup-asvMQIqTGDHm:policyName/awseb-e-g39cncr8-stack-AWSEBAutoScalingScaleUpPolicy-67CCjEMHn90
March 28, 2026 12:54:26 (UTC+5:30)	INFO	Created Auto Scaling group policy named: arn:aws:autoscaling:us-east-1:352917280532:scalingPolicy:41a459a6-659d-4d78-a0fb-92cd0f52058:autoScalingGroupName/awseb-e-g39cncr8-stack-AWSEBAutoScalingGroup-asvMQIqTGDHm:policyName/awseb-e-g39cncr8-stack-AWSEBAutoScalingScaleDownPolicy-fc5JBKq1JCh
March 28, 2026 12:54:25 (UTC+5:30)	INFO	Waiting for EC2 instances to launch. This may take a few minutes.
March 28, 2026 12:54:25 (UTC+5:30)	INFO	Created Auto Scaling group named: awseb-e-g39cncr8-stack-AWSEBAutoScalingGroup-asvMQIqTGDHm
March 28, 2026 12:54:03 (UTC+5:30)	INFO	Environment health has transitioned to Pending. Initialization in progress (running for 15 seconds). There are no instances.
March 28, 2026 12:53:55 (UTC+5:30)	INFO	Created security group named: sg-051e1c61ed44ab03c
March 28, 2026 12:53:55 (UTC+5:30)	INFO	Created target group named: arn:aws:elasticloadbalancing:us-east-1:352917280532:targetgroup/awseb-AWSEBYCSROTE9T8MB/44b2e79d2343dd39
March 28, 2026 12:53:39 (UTC+5:30)	INFO	Created security group named: sg-0b4b5640bed4ed76f
March 28, 2026 12:53:18 (UTC+5:30)	INFO	Using elasticbeanstalk-us-east-1-352917280532 as Amazon S3 storage bucket for environment data.
March 28, 2026 12:53:16 (UTC+5:30)	INFO	createEnvironment is starting.

us-east-1.console.aws.amazon.com / Environment overview - events Elastic Beanstalk us-east-1

Elastic Beanstalk > Environments > MyApp-env-1

Environment successfully launched.

Events (14)

Time	Type	Details
March 28, 2026 12:56:11 (UTC+5:30)	INFO	Successfully launched environment: MyApp-env-1
March 28, 2026 12:56:08 (UTC+5:30)	INFO	Application available at MyApp-env-1.eba-qddh25cp.us-east-1.elasticbeanstalk.com.
March 28, 2026 12:56:03 (UTC+5:30)	INFO	Added instance (i-0286287438f37e659) to your environment.
March 28, 2026 12:55:55 (UTC+5:30)	INFO	Instance deployment completed successfully.
March 28, 2026 12:55:45 (UTC+5:30)	INFO	Created Load Balancer listener named: arn:aws:elasticloadbalancing:us-east-1:352917280532:listener/app/awseb-NvpYdZ0rZ4hw/b6f91e613a7de9f1/bb0ef78b1bb694ba
March 28, 2026 12:55:42 (UTC+5:30)	INFO	Created load balancer named: arn:aws:elasticloadbalancing:us-east-1:352917280532:loadbalancer/app/awseb-NvpYdZ0rZ4hw/b6f91e613a7de9f1
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March 28, 2026 12:54:26 (UTC+5:30)	INFO	Created Auto Scaling group policy named: arn:aws:autoscaling:us-east-1:352917280532:scalingPolicy:709a26b7-4df6-4964-a4e6-93c74e19f3c:autoScalingGroupName/awseb-e-g39cncr8-stack-AWSEBAutoScalingGroup-asvMQIqTGDHm:policyName/awseb-e-g39cncr8-stack-AWSEBAutoScalingScaleUpPolicy-67CCjEMHn90
March 28, 2026 12:54:26 (UTC+5:30)	INFO	Created Auto Scaling group policy named: arn:aws:autoscaling:us-east-1:352917280532:scalingPolicy:41a459a6-659d-4d78-a0fb-92cd0f52058:autoScalingGroupName/awseb-e-g39cncr8-stack-AWSEBAutoScalingGroup-asvMQIqTGDHm:policyName/awseb-e-g39cncr8-stack-AWSEBAutoScalingScaleDownPolicy-fc5JBKq1JCh
March 28, 2026 12:54:25 (UTC+5:30)	INFO	Waiting for EC2 instances to launch. This may take a few minutes.
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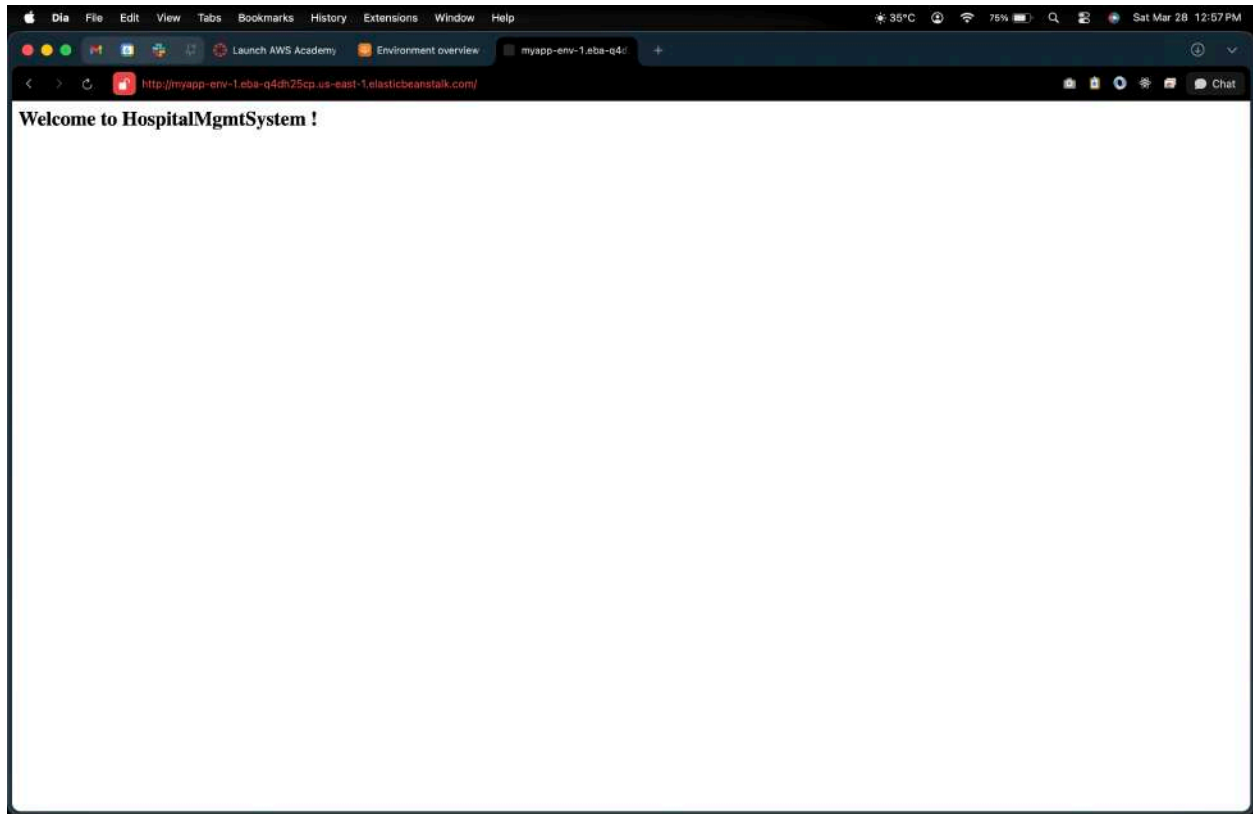
Access Your Application

- After deployment:
 - You will get a **URL (domain link)**
 - Open it in browser → Your app will run

The screenshot displays the AWS Elastic Beanstalk console interface. At the top, a green banner indicates 'Environment successfully launched.' Below this, the 'MyApp-env-1' environment overview is shown. The 'Health' status is 'Ok', and the 'Domain' is 'MyApp-env-1-eba-q4d25cp-us-east-1.elasticbeanstalk.com'. The 'Platform' section shows 'Tomcat 11 with Corretto 25 running on 64bit Amazon Linux 2023/5.12.0'. The 'Running version' is 'label', and the 'Platform state' is 'Supported'. A notification bar states: 'We've added the Deployments tab to help you track and troubleshoot environment changes. Let us know what you think.' Below this, the 'Deployment history (1)' table shows a single deployment:

Request ID	Status	Type	Start time	Duration
6cb0b3ef-7060-499a-bcac-47774c5fb106	Succeeded	Environment Creation	March 28, 2026 12:53:16 (UTC+5:30)	2m 54s

The footer of the console shows '© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences'.



Updating Application

1. Go to your environment
2. Click **Upload and Deploy**
3. Upload new version

